

Speech Processing (EC4143)

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Module 1

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Module 1:

Fundamentals of Speech Production and Phonetics

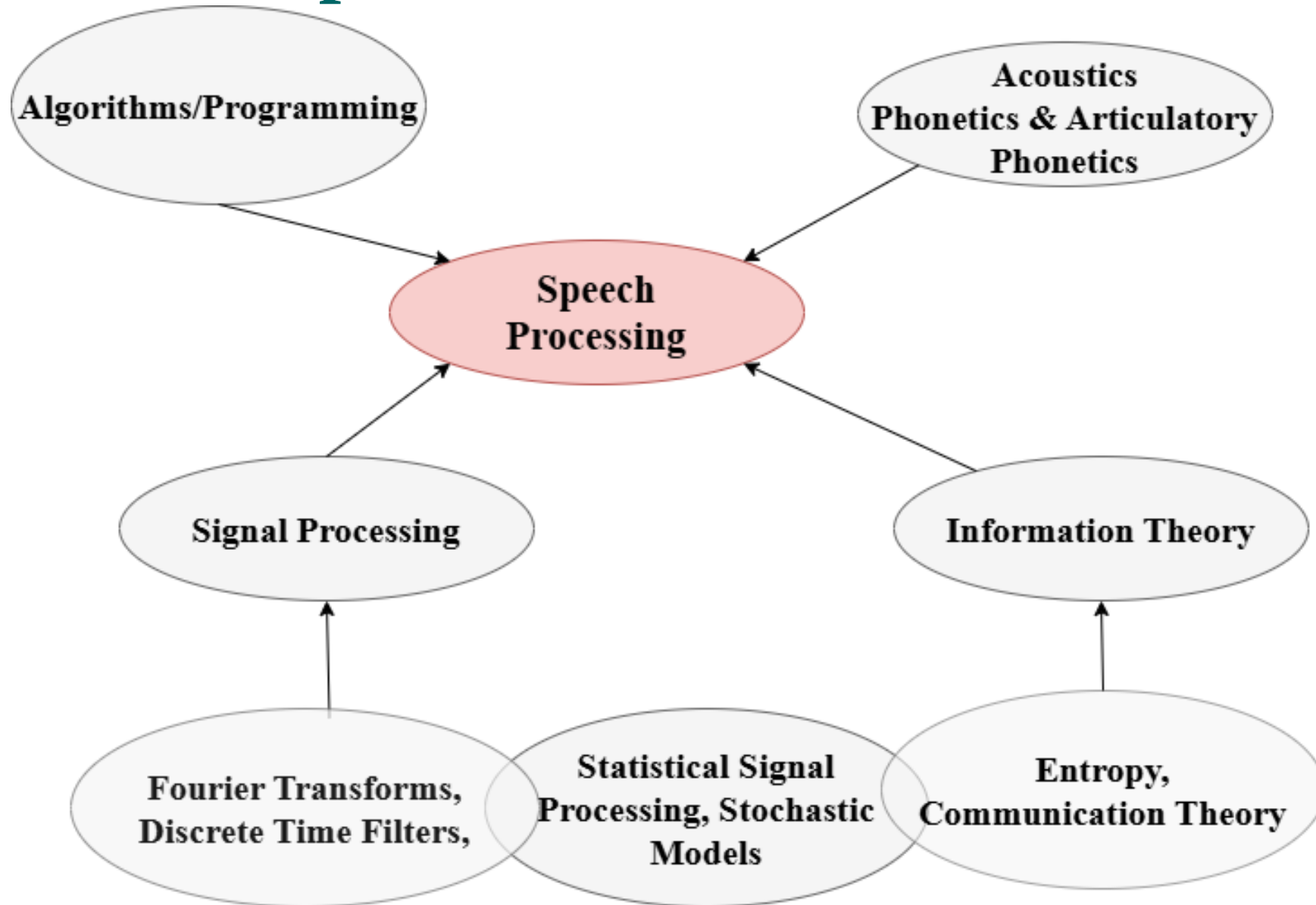
- Human speech production mechanism
- Source–filter model of speech
- Categorization of Speech
- Elements of a language: Phonemes in American language
- Place and Manner of Articulation
- Acoustic Characteristics of Speech

Speech Processing Applications

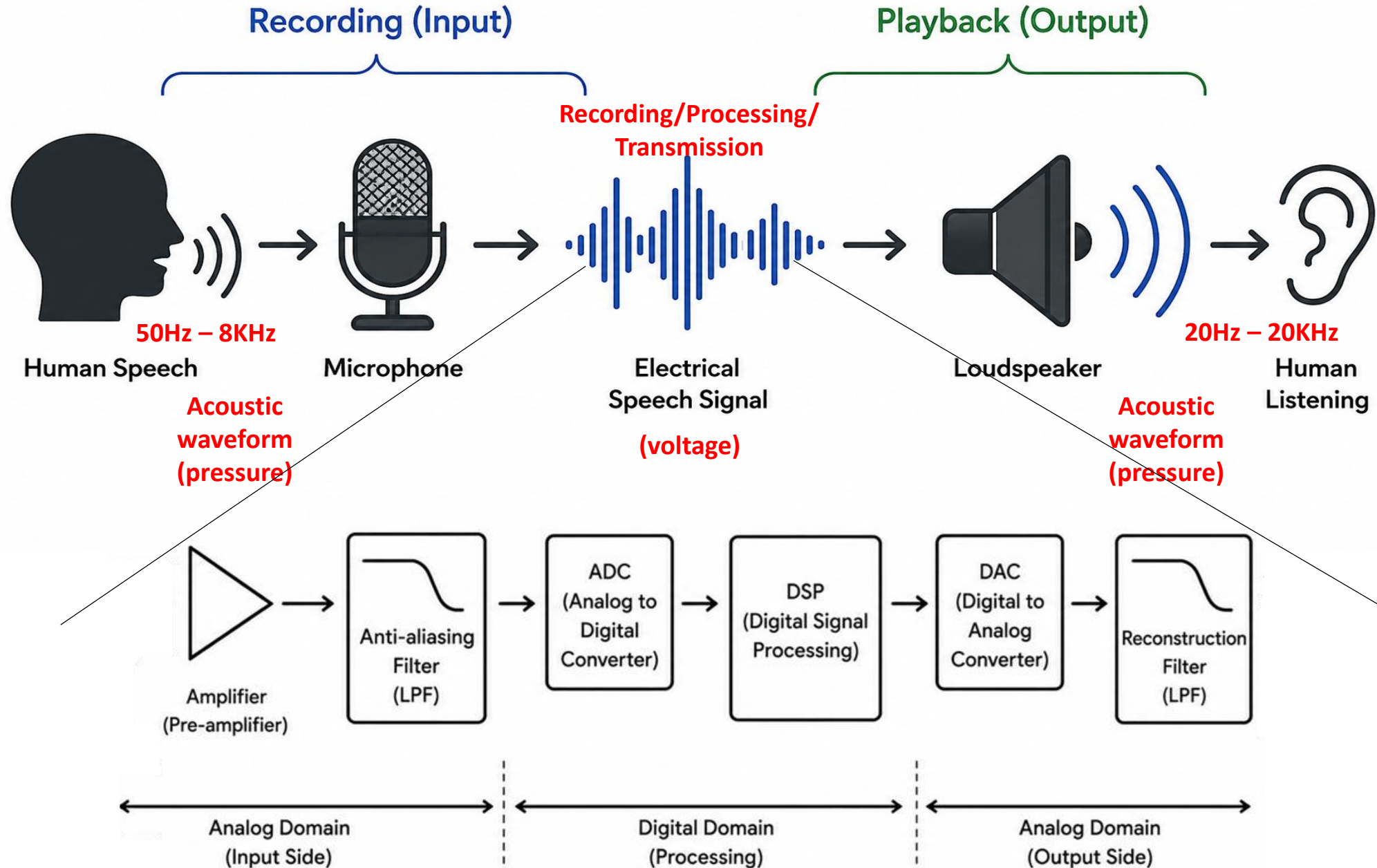
- 1. Speech Recognition:** Speech → Text/Command
Ex: Voice assistants, voice typing, call centres
- 2. Speaker Recognition:** Voice → Speaker Identity
Ex: Voice authentication, banking, security
- 3. Automatic Language Translation:** Speech → Text/Speech
Ex: Live translation, meetings, travel
- 4. Speech Enhancement:** Noisy Speech → Clean Speech
Ex: Mobile phones, hearing aids, online meetings
- 5. Text-to-Speech:** Text → Speech
Ex: Navigation, screen readers, virtual assistants

List some of the other speech processing applications with examples.

Different Disciplines of SP



Speech Production & Perception



DSP Concepts:

1. Sampling and Quantization: Convert analog signal to digital signal
2. Classification of Signals:
 - a. Energy and Power Signal

Speech is a finite-duration bounded signal → it is an energy signal.
 - b. Periodic and Aperiodic Signal

Speech has both periodic (voiced) and aperiodic (unvoiced) elements.
 - c. Symmetric and Asymmetric Signal

Speech is asymmetric signal.
 - c. Stationary and Non-stationary Signal

Speech is a non-stationary signal.

DSP Concepts (Contd.):

3. Classification of Systems: (System respond to the signal)

a. Static/memoryless and Dynamic System:

Vocal Tract is dynamic

b. Time varying and Time invariant System

Vocal Tract is time-varying

c. Linear and Non-linear System

Vocal Tract is linear

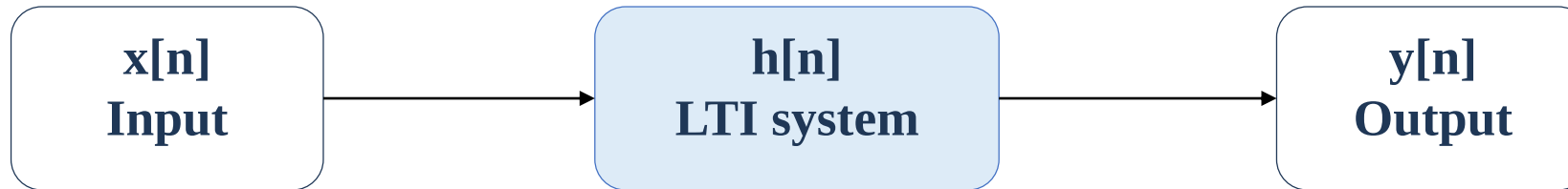
d. Causal and Non-causal System

Vocal Tract is causal

e. Stable and Unstable System

Vocal Tract is stable

DSP Concepts: LTI System



$$y[n] = h[n] * x[n]$$

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

$h[n]$ = impulse response of the system

Let $x[n] = e^{j\omega n}$

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]e^{j\omega(n-k)}$$

$$\begin{aligned} &= \sum_{k=-\infty}^{\infty} h[k]e^{j\omega n}e^{-j\omega k} \\ &= e^{j\omega n} \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k} \end{aligned}$$

$$y[n] = e^{j\omega n} H(e^{j\omega})$$

$H(e^{j\omega})$ is the frequency response of the system

DSP Concepts: LTI System (Contd.)

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h[k] e^{-j\omega k}$$

$e^{-j\omega k} = \cos(\omega k) - j\sin(\omega k) \rightarrow \text{complex}$

$$H(e^{j\omega}) = \underbrace{\sum_{k=-\infty}^{\infty} h[k] \cos(\omega k)}_{\text{Real}} + \underbrace{j \sum_{k=-\infty}^{\infty} h[k] \sin(-\omega k)}_{\text{Imaginary}}$$

Magnitude response

$$|H(e^{j\omega})| = \sqrt{\text{Re}^2\{H\} + \text{Im}^2\{H\}}$$

how much each frequency is amplified or attenuated

Phase response

$$\begin{aligned} \varphi(\omega) &= \arg[H(e^{j\omega})] \\ &= \tan^{-1}(\text{Im}\{H\}/\text{Re}\{H\}) \end{aligned}$$

how the phase of each frequency component is shifted

Group delay

$$\tau(\omega) = -d\varphi(\omega)/d\omega$$

frequency-dependent delay introduced by the system

Introduction

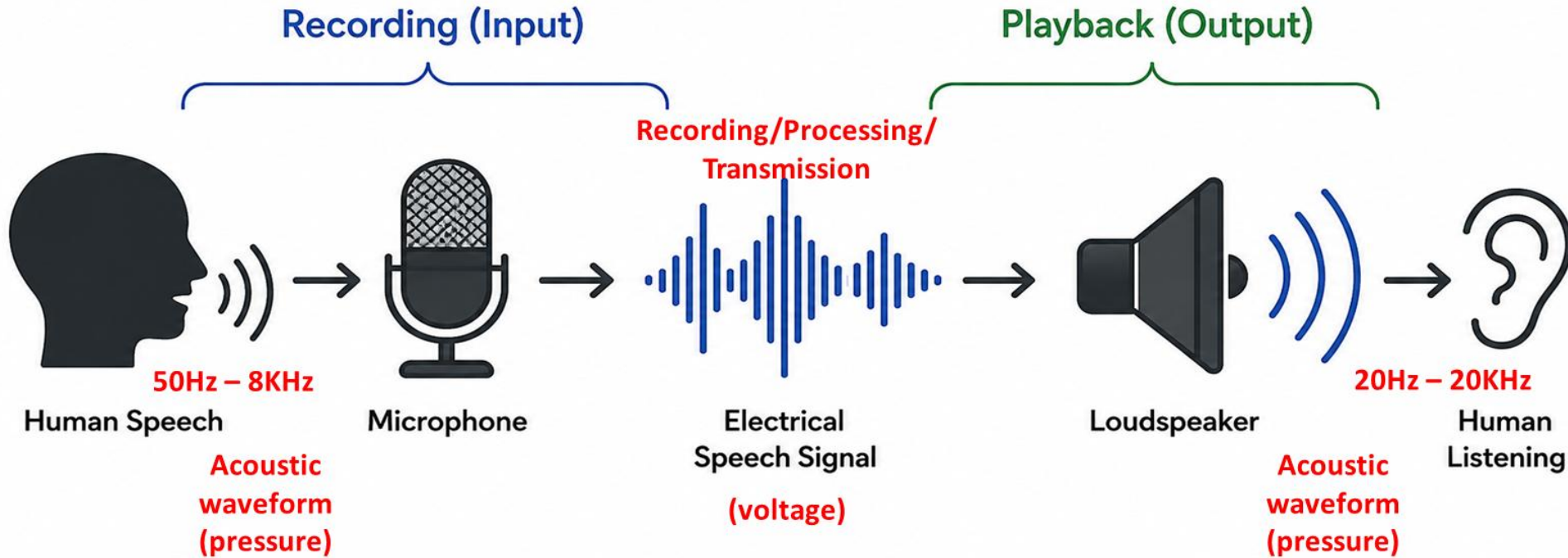


Fig 1: Human Speech Production and Perception

Introduction (Contd.)

- The fundamental purpose of speech is communication.
- It is produced by the vibration of the air through the vocal folds.
- It is a continuously varying acoustic signal.
- Microphone converts this acoustic signal to electrical and the loudspeaker converts it back to the acoustic form.
- This form of speech processing is the basis for from Bell's telephone as well as today's devices for recording/transmission/manipulation.
- Every speaker has unique speech production characteristics.
- Every speech has a sequence of sounds called phonemes.

Introduction (Contd.)

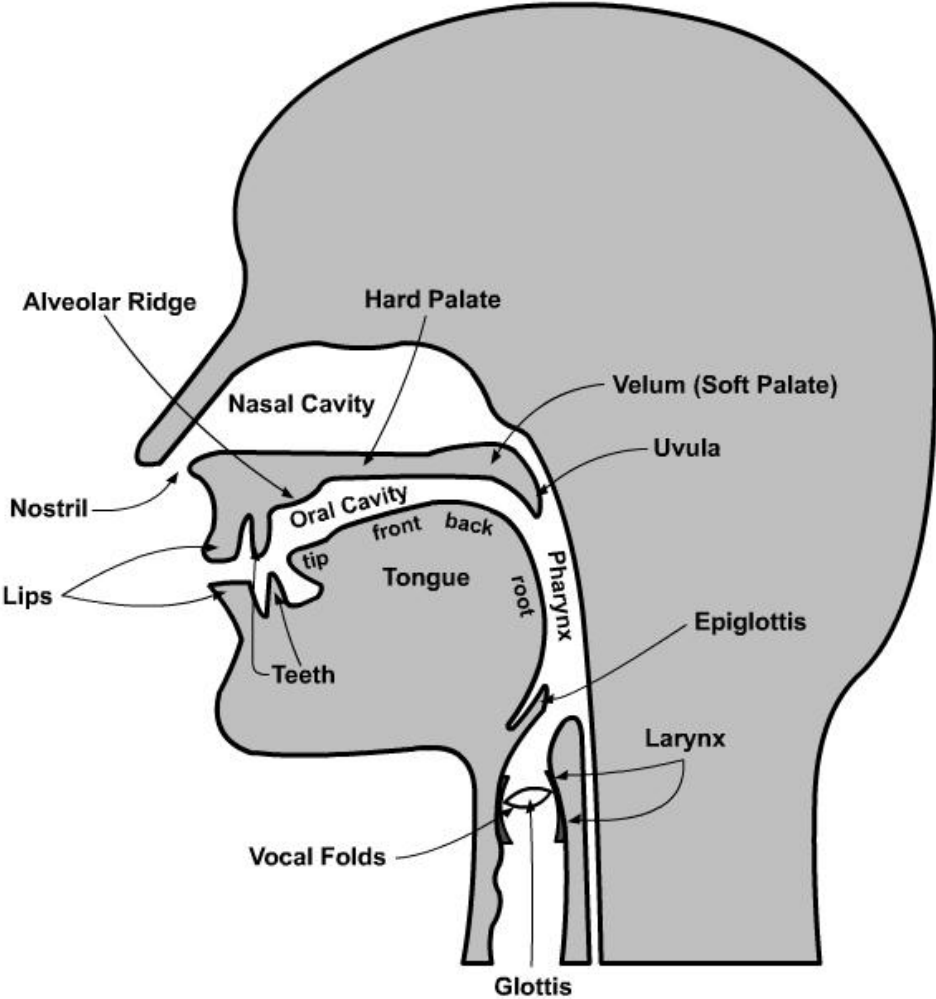


Fig 2: Human Speech Organs

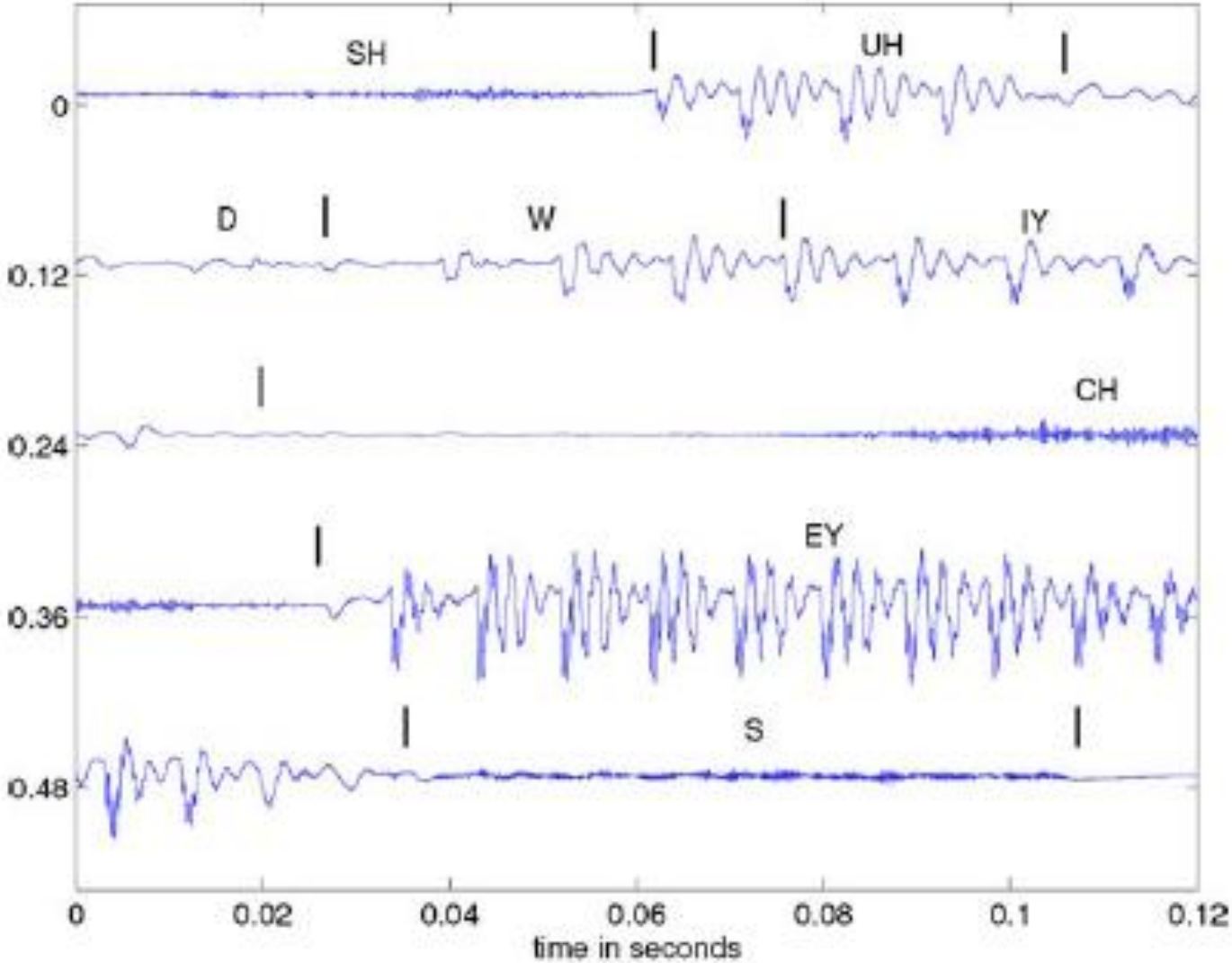


Fig 3: Example of a speech waveform with phonetic labels

Speech Chain

- Speech chain is the process of producing and perceiving speech.
- It involves:
 - a. Formulation of the message in the brain of a talker.
 - b. Creation of the speech signal.
 - c. Understanding the speech by a listener.
- Generally has three stages:
 - a. Speech Production
 - b. Transmission Channel
 - c. Speech Perception

Speech Chain (Contd.)

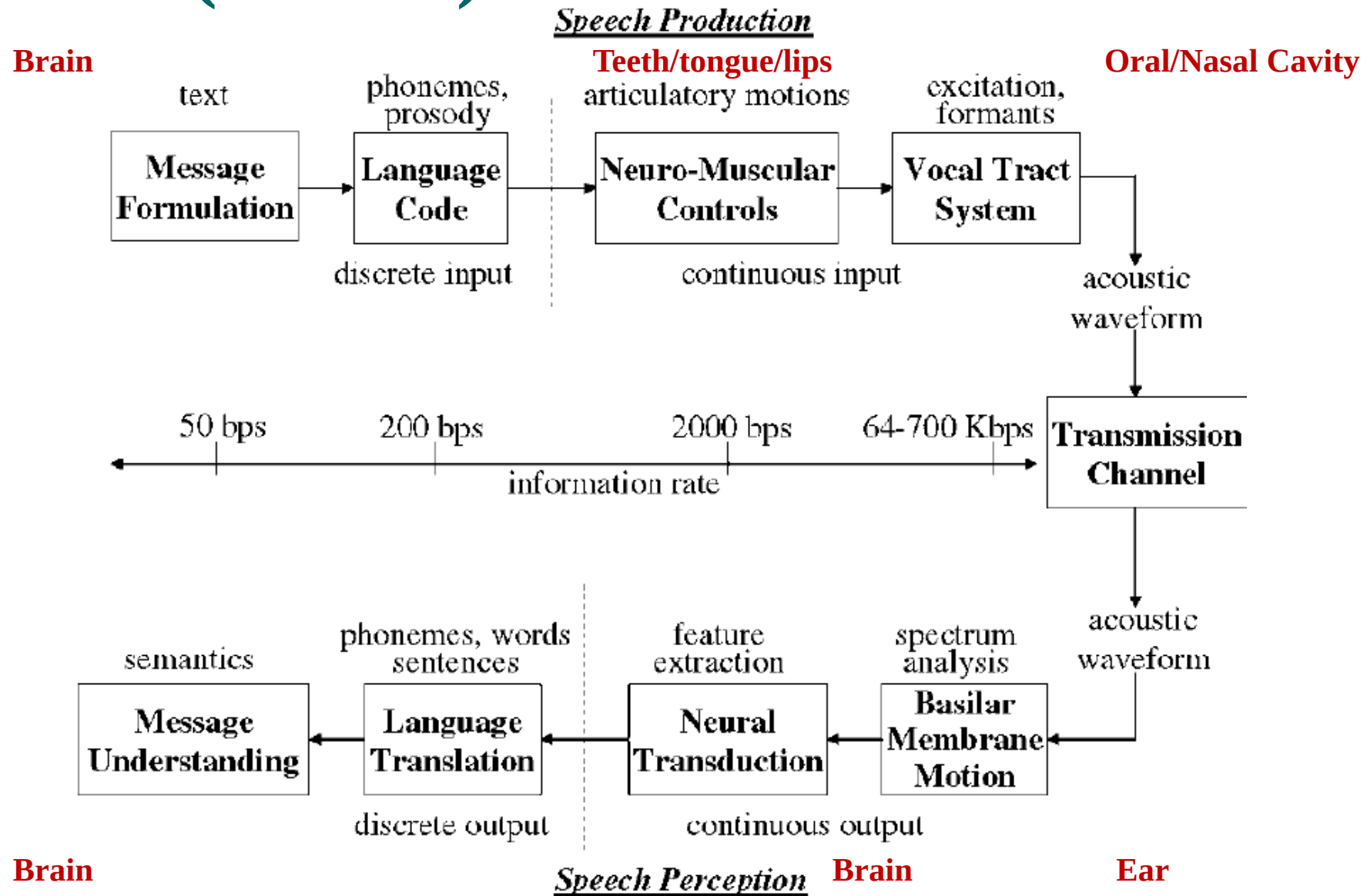


Fig 3: Speech Chain

Speech Signal

- Speech signal is a sequence of sounds.
- Each sound is referred as phoneme. (Phonemes are not alphabets.)
- Phoneme – Fundamental unit of speech.
- Phonemes are arranged in a specific language rule.
- Linguistics: The study of language rules.
- Phonetics: The study of different phonemes.
- In most languages, the #phonemes ranges b/w 32 to 64.
- Each phoneme is represented b/w 2 slashes: (cat → /K/ /AE/ /T/).
- Replacing one phoneme changes the meaning: (bat → /B/ /AE/ /T/).

Speech Signal (Contd.)

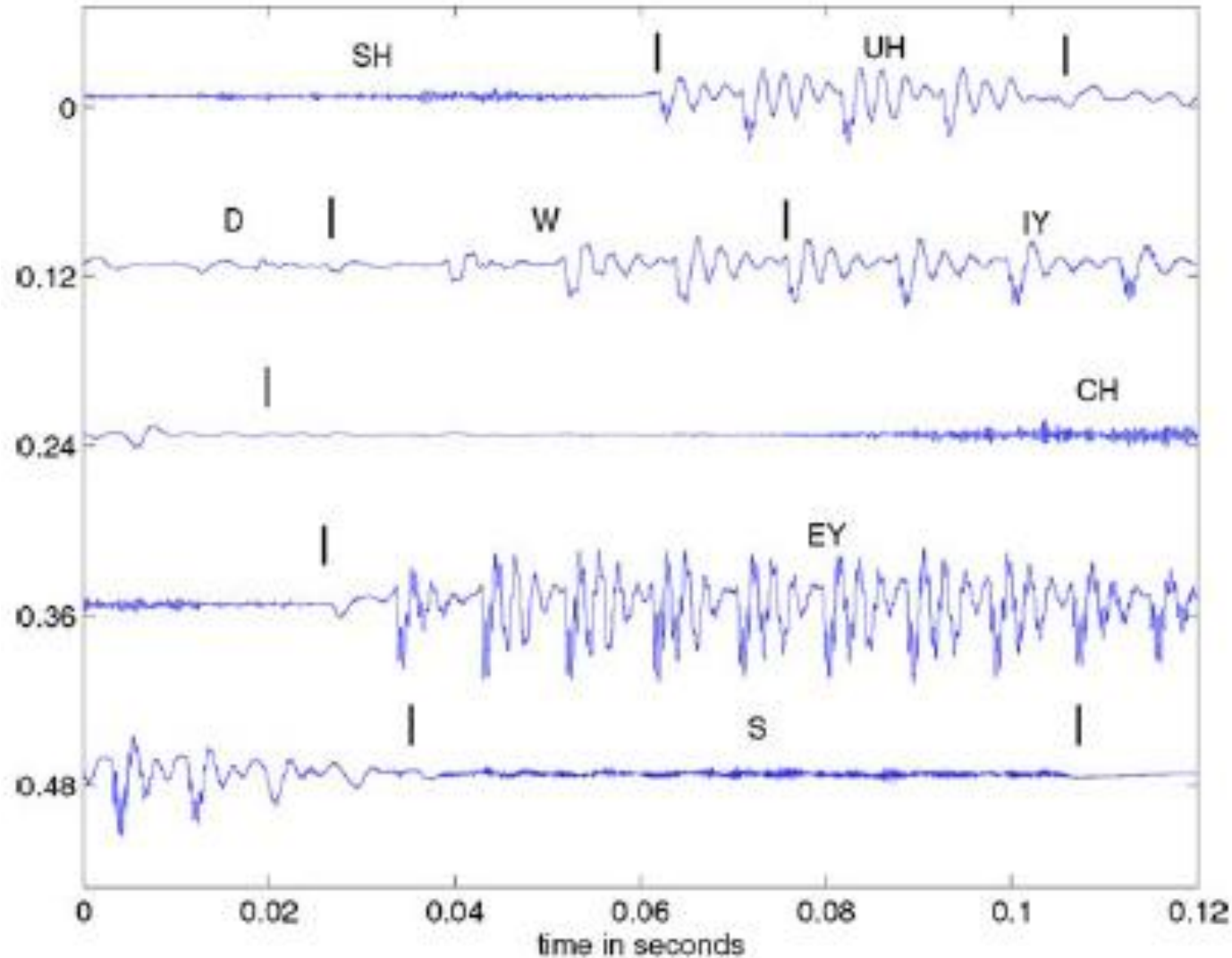


Fig 4: Example of a speech waveform with phonetic labels for the message “Should we chase”

- #alphabets = 13
- #phonemes = 8
- ARPAbet – computer-keyword friendly code.
- /SH/, /S/ → unstructured and noisy.
- /UH/, /IY/, /EY/ → structured and quasi-periodic.

Human Speech Production

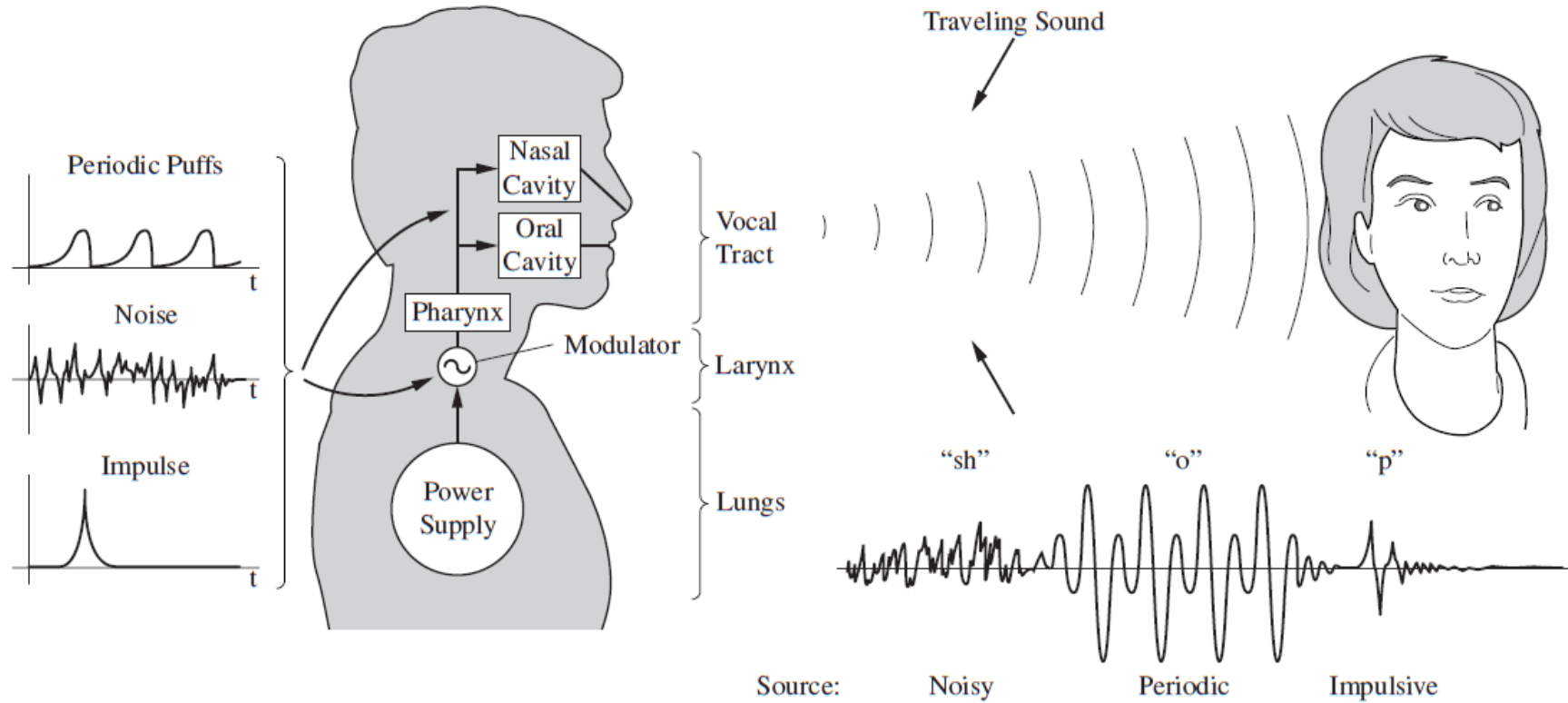
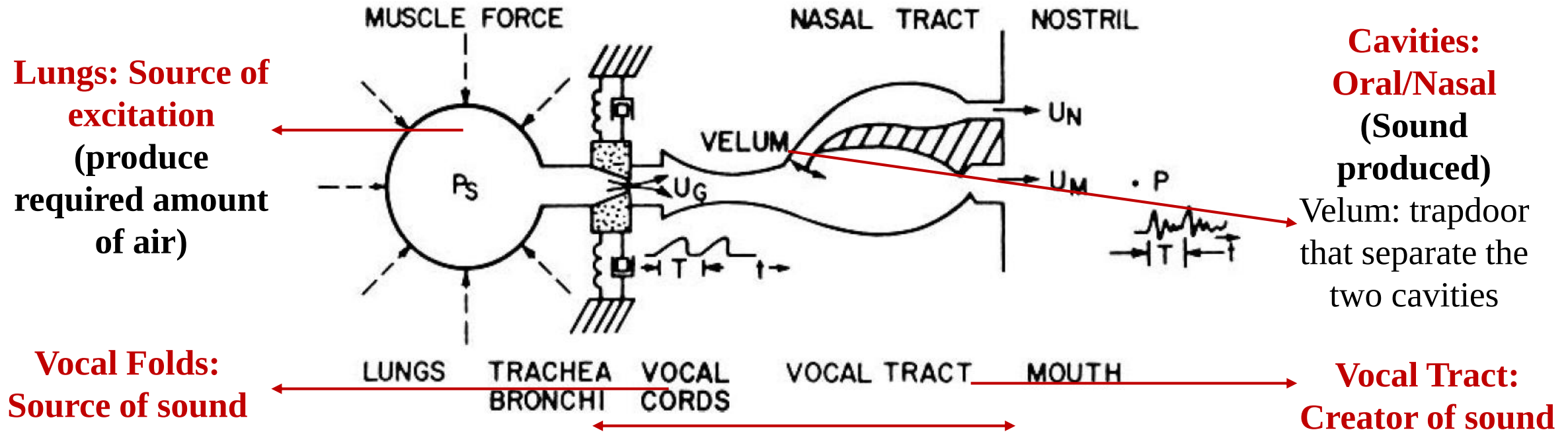


Fig 5: Simple View of Human Speech Production

Main speech organs involved in speech production

- **Lungs**- provide airflow to the larynx
- **Larynx**- modulates airflow from the lungs and provides puff-like or noisy airflow source to the vocal tract
- **Vocal tract** – modulates the airflow from the larynx by shaping the spectrum of the source

Schematic Representation of Human Speech Production System



Vocal Tract: Mechanical body (produce vibration and creates sound)

It is a tube of non-uniform cross-sectional area bounded by vocal folds and oral cavity at two ends.

- Vocal Folds/Cords:** A pair of muscle in the voice box (larynx). Air from the lungs are vibrated in the vocal folds.
Open – no vibration (no sound), **Close** – vibration of air.
- Articulators:** hard palate, tongue, teeth, lips, etc. The shape of the vocal tract changes based on the motion of the articulators. (Vibration + Vocal tract shape → sound/phoneme)

Categories of Sound

I. Based on the Vocal Fold Vibration:

- i. **Voiced:** Vocal folds → open and close (vibration). The vocal tract tube is excited by the pulses of air pressure resulting from quasi-periodic opening and closing of vocal folds. Ex: Vowels /UH/ /IY/ , liquids /W/ /L/, glides /Y/ /R/, nasals /M/ /N/.
- ii. **Unvoiced:** Vocal folds are slightly opened (no vibration). Creates constriction somewhere in the vocal tract tube and forces the air pass through that constriction, causing turbulent air airflow. (random noise).
Ex: /SH/ /S/.
- iii. **Silence:** Vocal folds are fully opened (no vibration)

Categories of Sound (Contd.)

II. Based on the Constriction of the Vocal Tract Tube:

i. **Stops:** Tube is fully closed, air burst through the oral cavity.

Ex: /K/ /P/ /T/

ii. **Fricatives:** Tube is partially closed (narrow constriction).

Ex: /F/ /V/ /TH/ /DH/ /S/ /Z/

iii. **Affricates:** Tube is fully closed for sometime, and partially opened

later. Ex: /CH/, /JH/

Note: Fricatives, Affricates, and Stops can be Voiced or Unvoiced based on the vibration of the vocal folds.

Activity

1. Categorize the below phonemes into voiced or unvoiced sounds:

/B/ /D/ /G/ /V/ /DH/ /Z/ /JH/

/P/ /T/ /K/ /F/ /TH/ /S/ /CH/

Voiced	Unvoiced	
/B/	/P/	 Fricatives
/D/	/T/	
/G/	/K/	
/V/	/F/	
/DH/	/TH/	
/Z/	/S/	 Affricates
/JH/	/CH/	

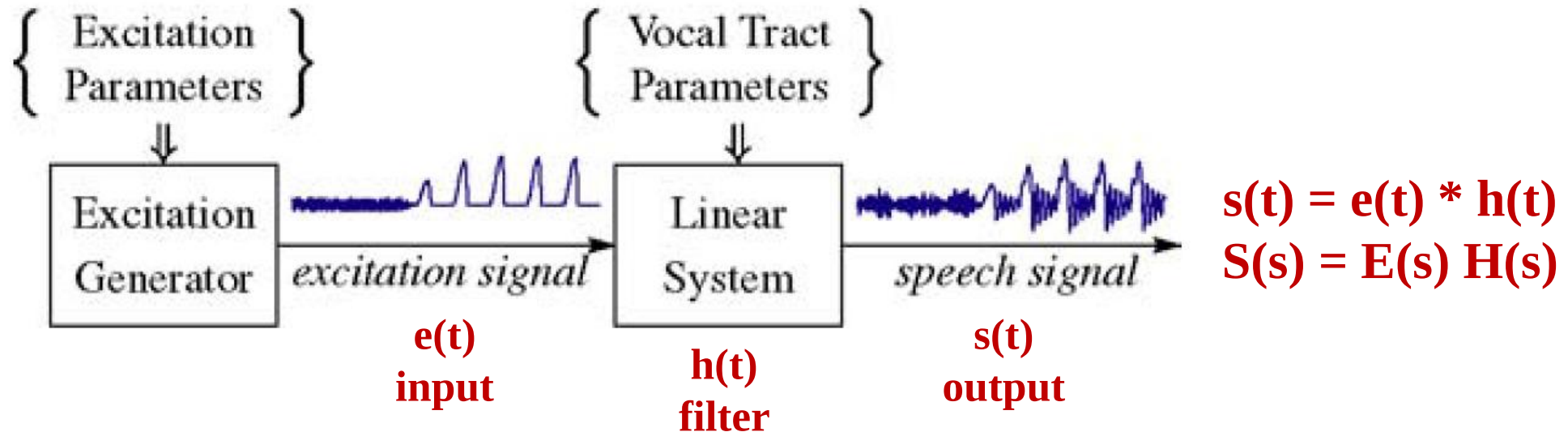
Practical

1. Speech Recording and Waveform Visualization:

- i. MATLAB code to record the speech signal.
- ii. Plot the waveform and categorize into different sounds (demarcations).

Source-Filter Model

- Engineering model of the human speech production system.

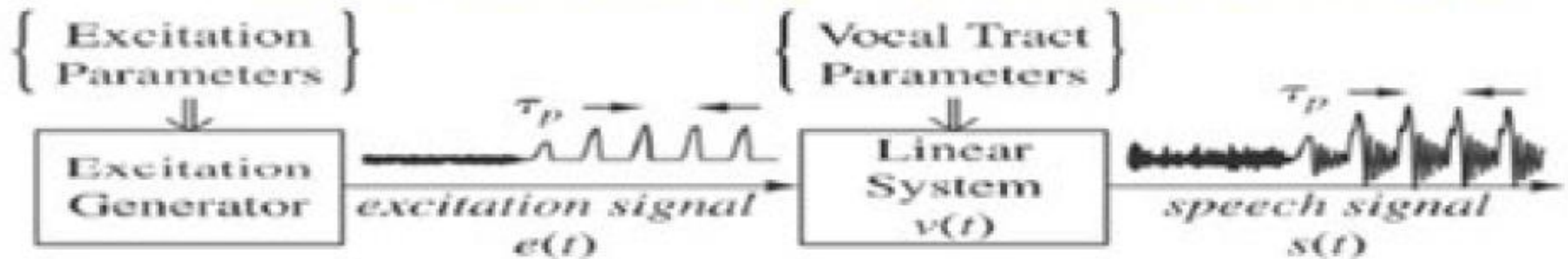


- Input** has different frequencies + fundamental frequency (**pitch**).
- The frequency response of the **vocal tract** modifies the input signal based on the positioning of the articulators, which are the instruments in forming the phonemes.
- Formant** frequencies – the resonance frequencies resulting from a particular configuration of the articulators.

Source-Filter Model (Contd.)

1. Source:

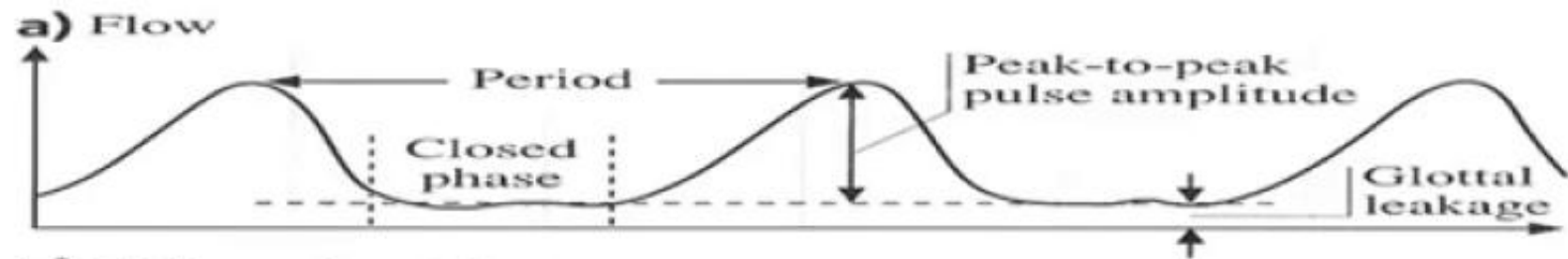
- Simulates different modes of sound generation in the vocal tract.
- Voiced $\rightarrow$ source $\rightarrow$ vocal folds vibrates $\rightarrow$ periodic pulses $\rightarrow$ pitch
- Unvoiced $\rightarrow$ source $\rightarrow$ vocal folds do not vibrate $\rightarrow$ noise $\rightarrow$ no pitch



Source-Filter Model (Contd.)

1. Source:

- Pitch/Fundamental frequency: frequency at which vocal folds vibrates.
- Every voiced sound has a specific pitch, denoted as 'F0'.
- In music, every note has a specific 'F0'.
- The average F0 range of a person is the characteristic feature of that person.



- Peak-to-Peak $\rightarrow$ 1 cycle.

$$avg\ t_0 = \frac{time}{No.of\ cycles} ; avg\ F_0 = \frac{1}{t_0}$$

Source-Filter Model (Contd.)

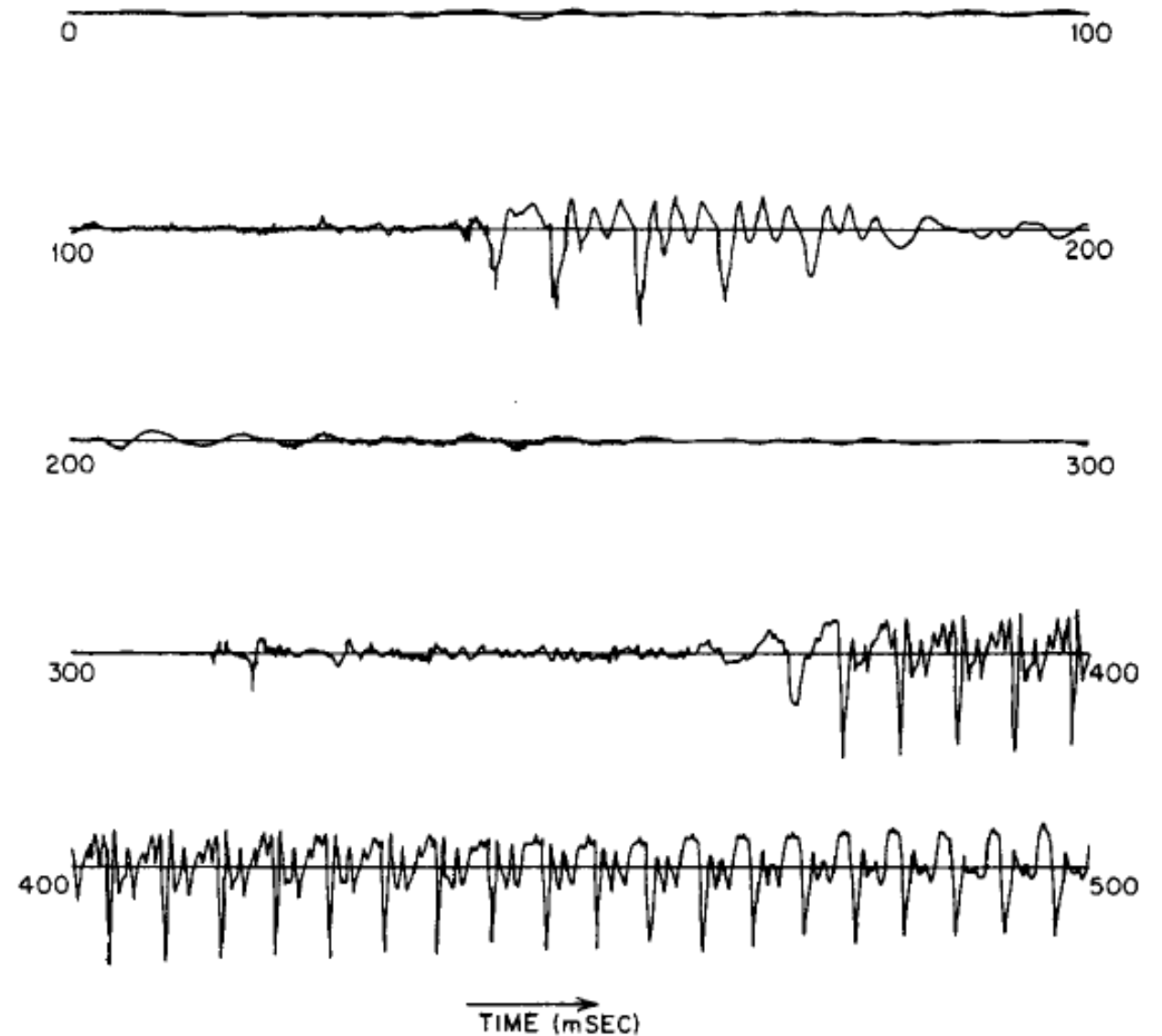
1. Source:

- The average F0 range changes between gender based on the length and mass of the vocal folds.
- Men: length ↑ and mass ↑, therefore avg F0 ↓
- Women: length ↓ and mass ↓, therefore avg F0 ↑
- **Children ?**

Gender	Vocal Fold Length	F0	Formants
Male	Long	Low (100 Hz – 150 Hz)	Lower
Women	Short	High (200 Hz – 250 Hz)	Higher
Children	Much shorter	Higher (300 Hz – 500 Hz)	Highest

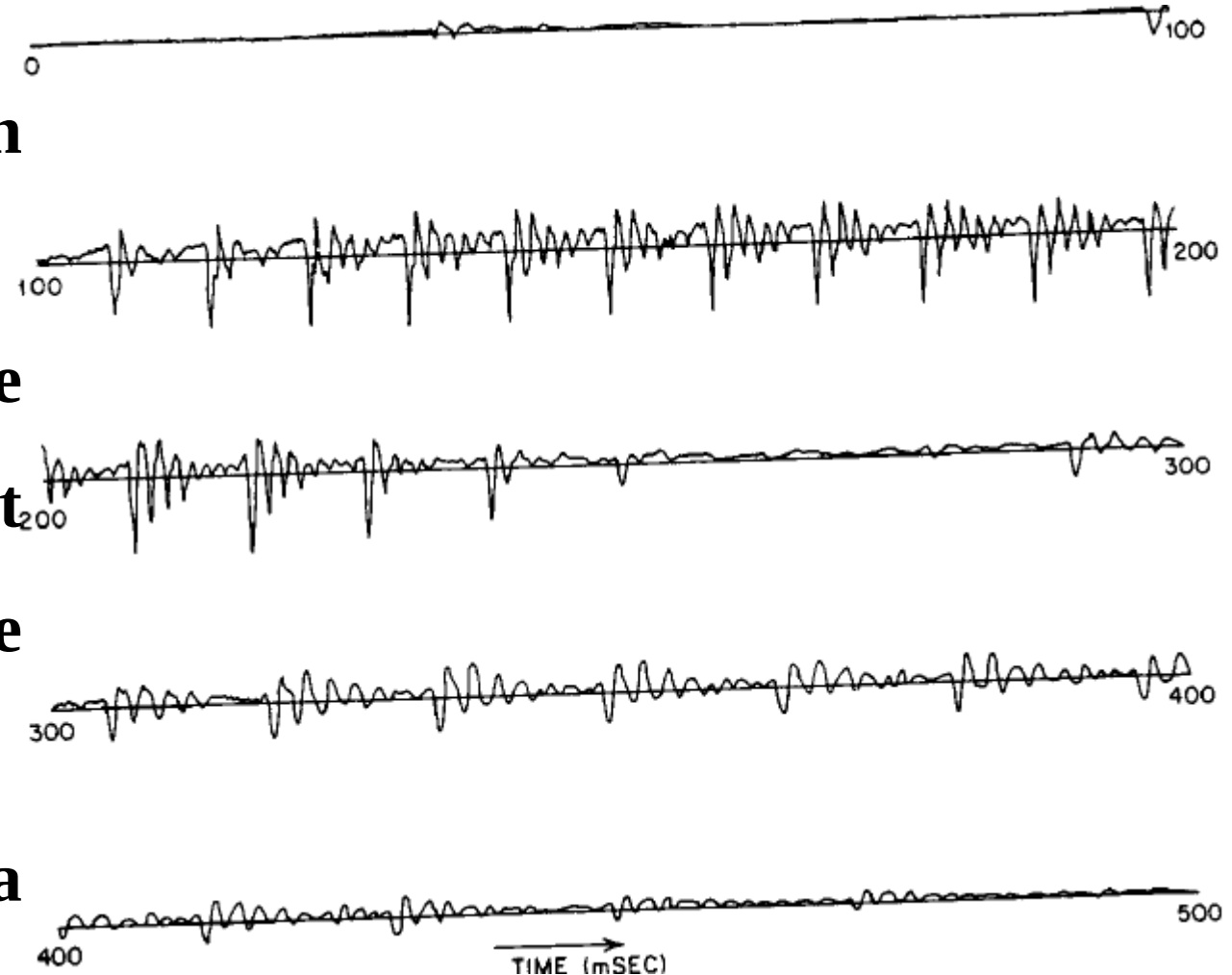
Problem 1

1. Indicate the voiced and unvoiced sounds
2. Find the average F0 of a voiced sound.
3. Identify the speaker.



Problem 2

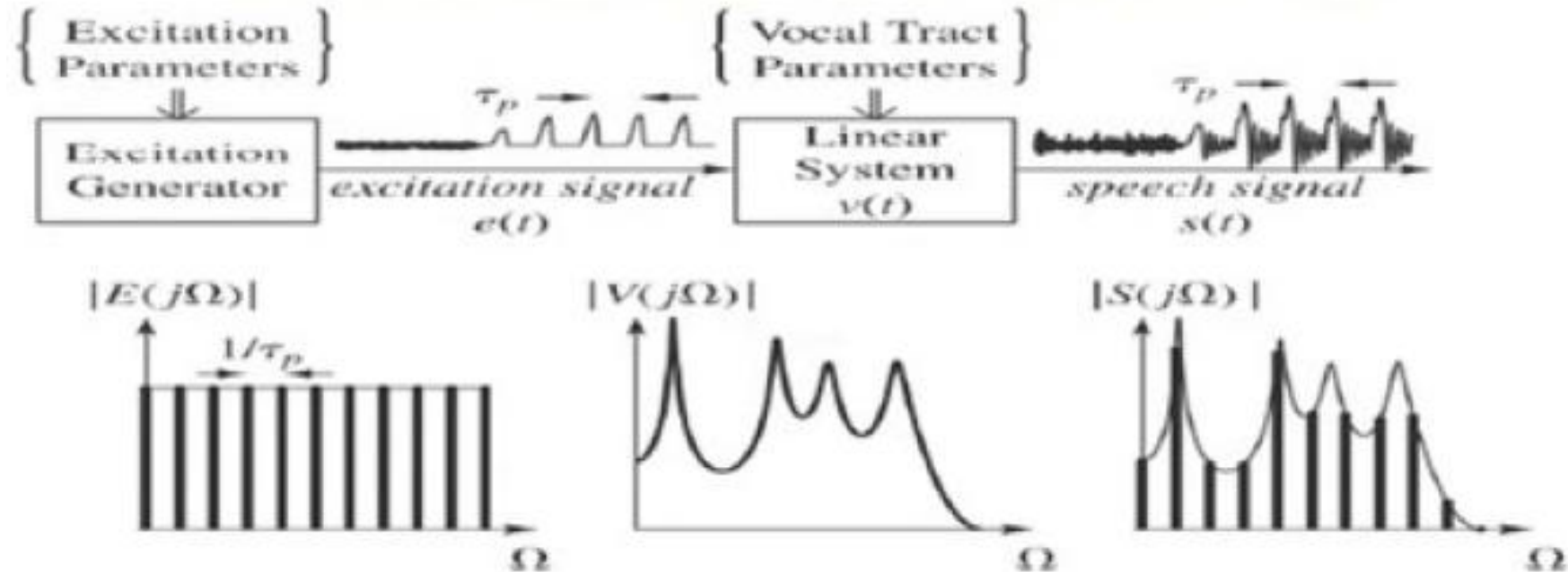
1. Indicate the boundaries between phonemes: /c/ /a/ /tt/ /le/
2. Indicate the point where the voice pitch frequency is (i) the highest and (ii) the lowest. What are the pitch frequencies at these points.
3. Is the speaker probably a male/female/child? How do you know?



Source-Filter Model (Contd.)

2. Filter: (LTI System)

- Alters the input excitation to produce recognizable phoneme.
- The articulators instrument the shape of the vocal tract emphasizes the formants.



Source-Filter Model (Contd.)

2. Filter: (LTI System)

- Analysed in frequency domain

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{1 - \sum_{k=1}^N a_k z^{-k}} = \frac{b_0 \prod_{k=1}^M (1 - d_k z^{-1})}{\prod_{k=1}^N (1 - c_k z^{-1})}$$

- Determining the poles $\rightarrow$ formants $\rightarrow$ sound/phoneme.
- Mathematically, formants $\rightarrow$ F1, F2, F3.
- Poles near to the unit circle $\rightarrow$ more amplification at the harmonics of F0 closer to the formants F1, F2, F3.
- Poles away from the unit circle $\rightarrow$ dispersed amplification at the harmonics of F0 closer and surrounding the formants F1, F2, F3.

Source-Filter Model (Contd.)

2. Filter: (LTI System)

- Consider a all-pole filter model:

$$V(z) = \frac{G}{1 - \sum_{k=1}^N a_k z^{-k}}$$

- The pole location in s-plane:

$$s_k, s_k^* = -\sigma_k \pm j2\pi F_k$$

- The pole location in z-plane:

$$z_k, z_k^* = e^{-\sigma_k T} e^{\pm j2\pi F_k T}$$

$$r = e^{-\sigma_k T}$$

$$\theta = \pm 2\pi F_k T$$

Source-Filter Model (Contd.)

Source	Fundamental Frequency (F0)	Source Property (Speaker)
Filter	Formant Frequencies (F1, F2, F3)	Filter Property (Phoneme)

- The speech signal can be represented by the parameters of the model.
- Analyse the model parameters over a short duration – time-invariant over short intervals - compute/process the speech signal – digital domain.

How to Describe a Sound?

1. **Manner of Articulation:** Partial/full air-flow.
2. **Place of Articulation:** Place of constriction/obstacle.

Manner of Articulation

1. **Stops** - Complete obstruction of airflow
2. **Fricatives** - Nearly complete obstruction of airflow
3. **Affricates** - stop followed by a fricative
4. **Nasals** - Air passes through nasal cavity
5. **Liquids** - Not narrow enough constriction
6. **Glides** - slight closure of the articulation

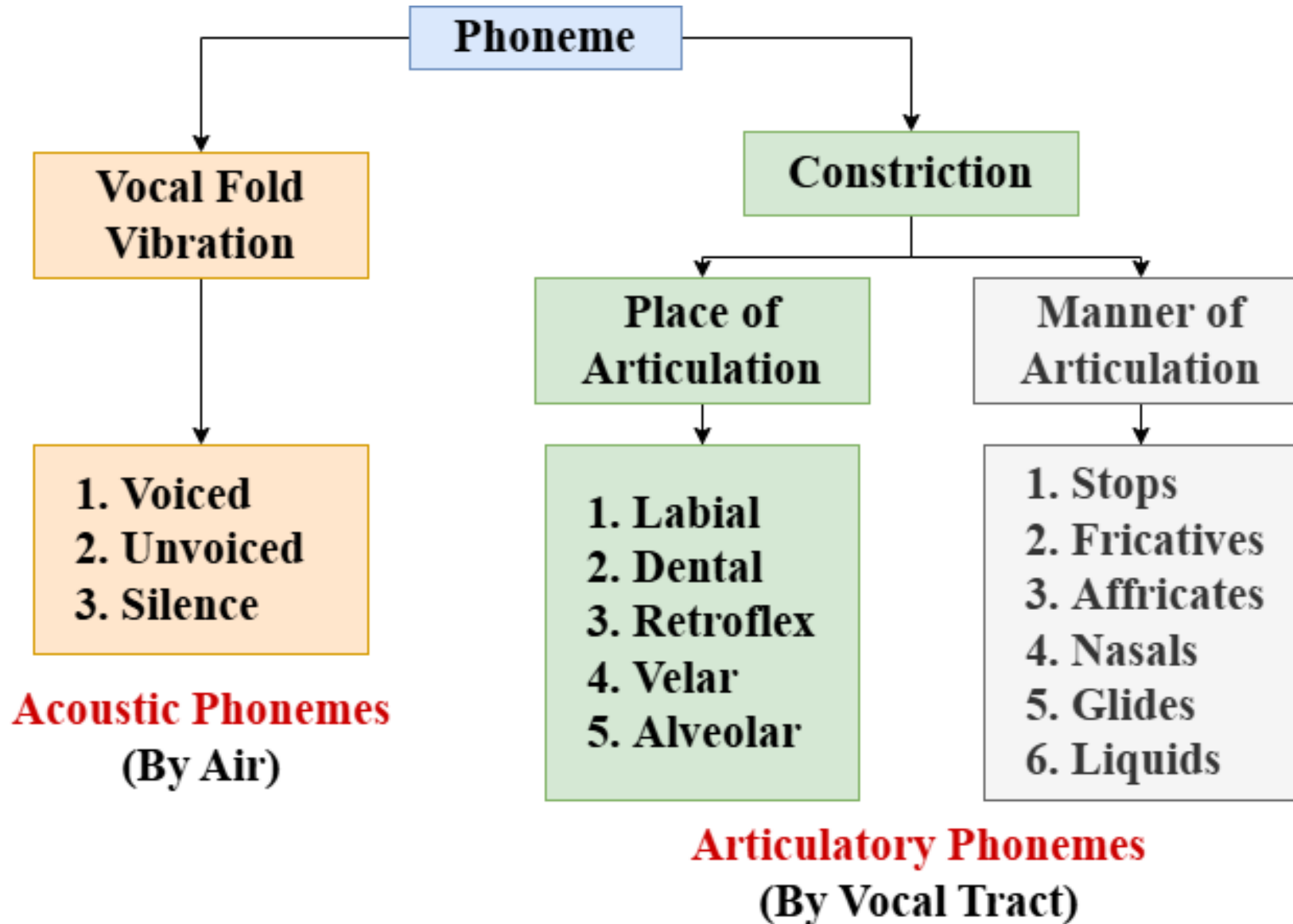
Manner of Articulation (Contd.)

Manner	English Phonemes	Hindi Phonemes
Stops	p,t,k,b,d,g	p, ṭ, t,k,b,ḍ,d,g ph,ṭh,th,kh,bh,ḍh,dh,gh
Fricatives	f,v,s,sh,z	s,sh
Affricates	dz,ch	ch,chh,zh,zhh
Nasals	m,n,ng	m,n,ng
Liquids	r,l	r,l
Glides	w,j	y,v

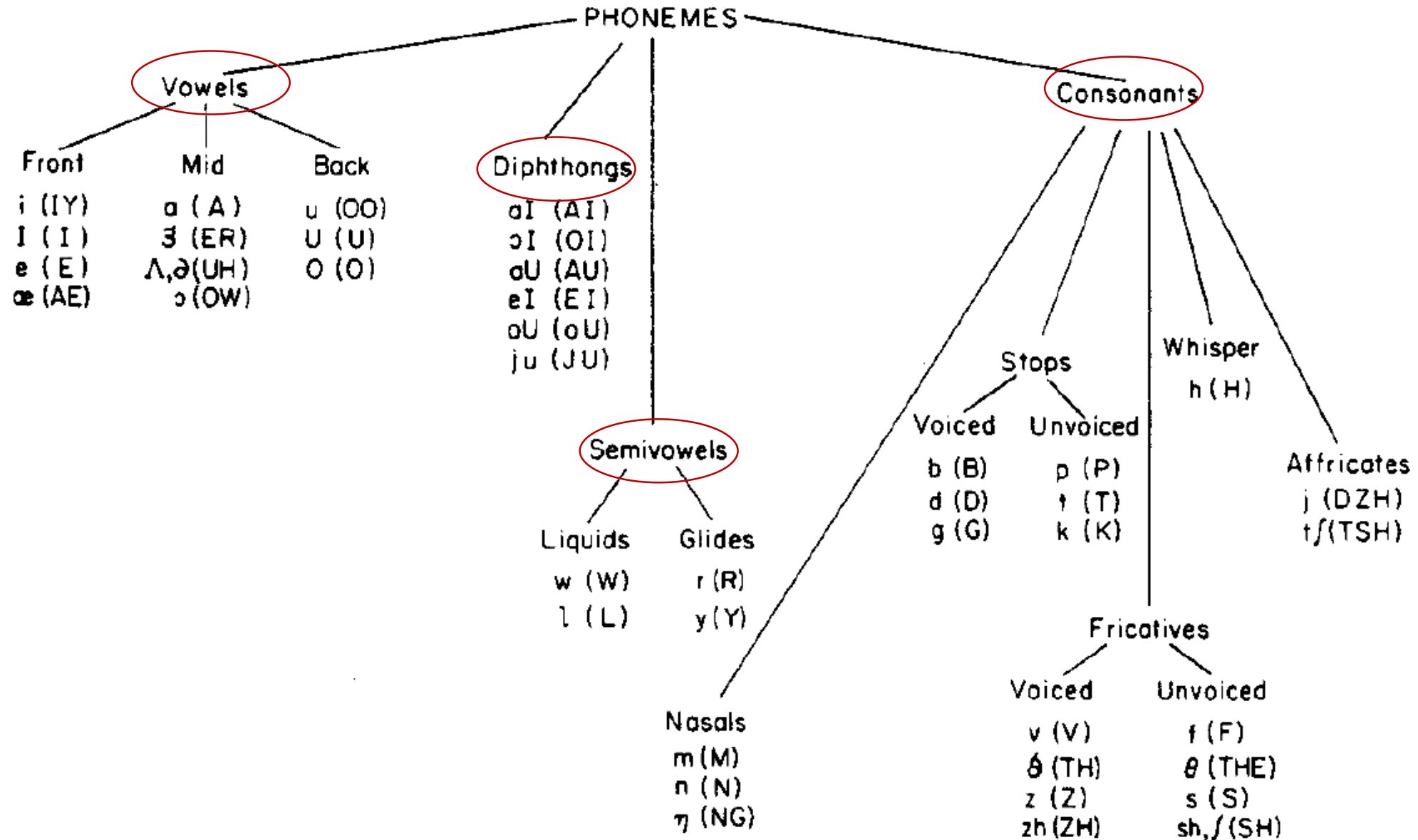
Place of Articulation

Place	English Phonemes (stops)	Hindi Phonemes (stops)
Labial	p, b,m	p,b,ph,bh
Dental		t̪,d̪
Alveolar	t,d	
Retroflex		ʈ,ɖ
Velar	k,g	k,kh,g,gh

Categories of Sound Phonetically



Categories of Sound Linguistically



Categories of Sound Linguistically (Contd.)

1. Vowels

- Largest phoneme
- No constriction of airflow
- Source – quasi-periodic air puffs
- Position of tongue + Roundness of lips = vowel
 - i. Position of the tongue → high/low and front/back
 - ii. Roundness of lips → rounded/unrounded
- Example:
 - i. /oo/ in food – high, back, rounded
 - ii. /I/ in thing – low, front, unrounded

Categories of Sound Linguistically (Contd.)

2. Diphthongs

- Gliding monosyllabic sound
- Start at one articulatory position and moves to another position
- Source – quasi-periodic air puffs
- Example: day, my, boy, house, air, pure

3. Semivowels

- Vowel like nature but function-like consonant
- Gliding transition between adjacent phonemes
- Source – quasi-periodic air puffs
- Example: yes, we, water

Note: Strengths of diphthongs, semivowels, and nasals are less than that of a vowel.

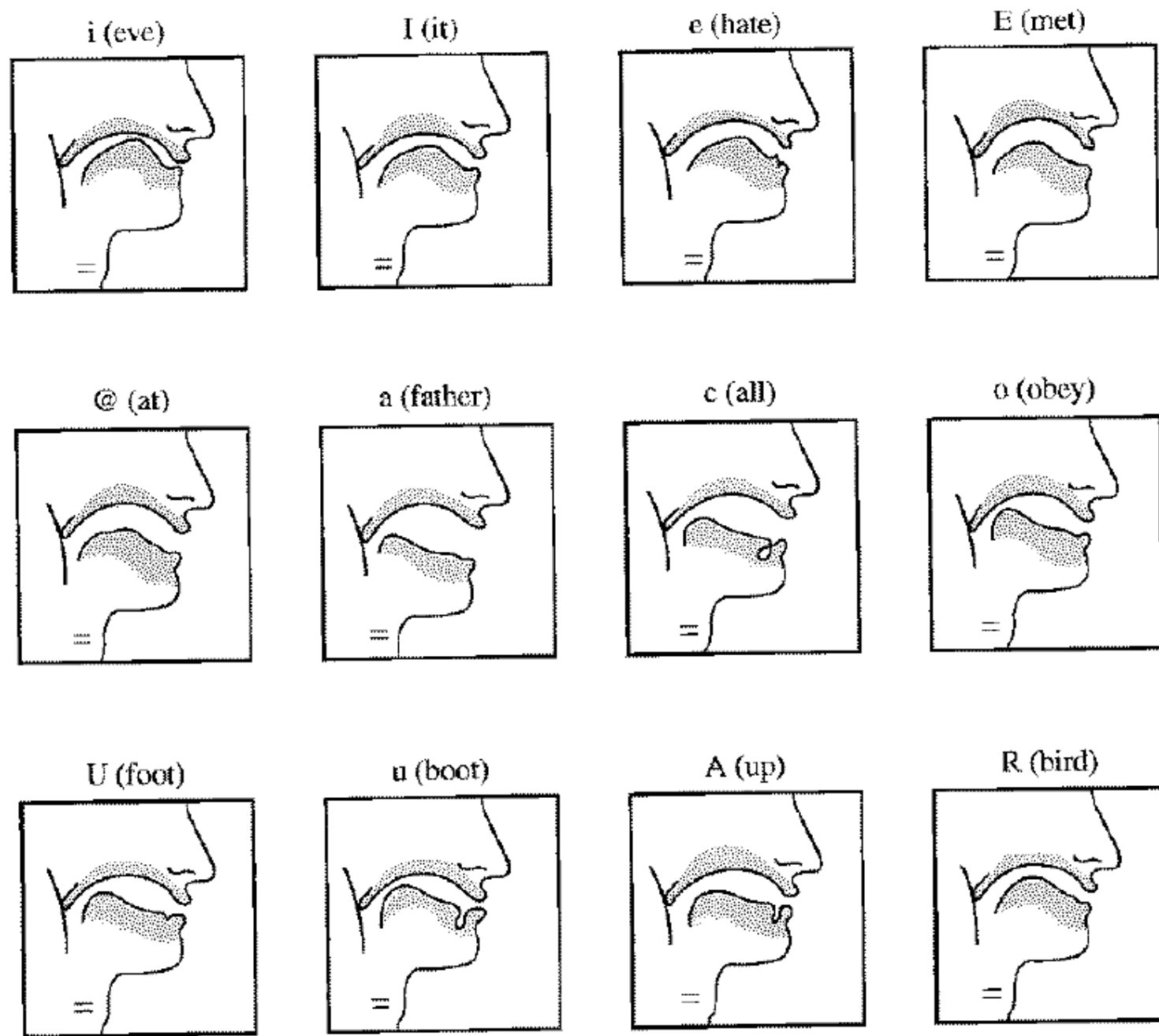


Fig 6: Vocal tract profiles for voiced sounds

Categories of Sound Linguistically (Contd.)

4. Consonants

- Second-largest phoneme
- Source - noise
- Airflow constriction
 - Fricatives** – Partial constriction
 - Plosives/Stops** – Total constriction → sudden release/burst
 - Nasal** – quasi-periodic air puffs, vocal folds vibrate, but constriction of air in the oral cavity, sound is radiated through the nostrils

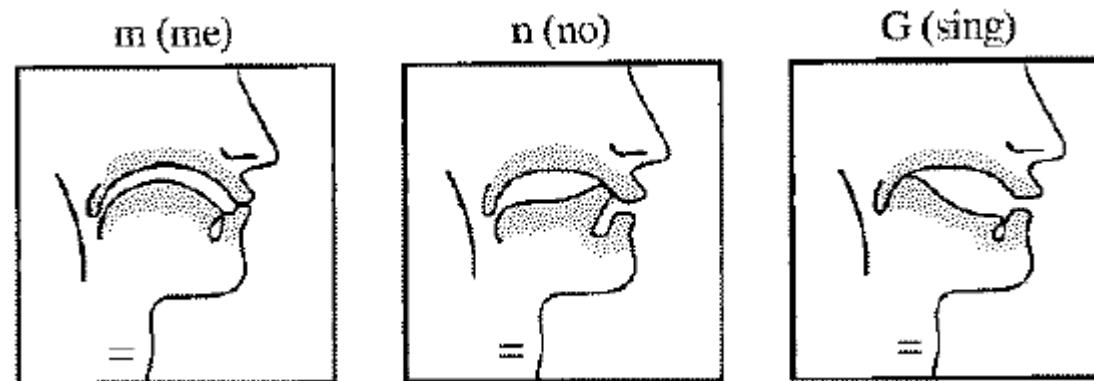


Fig 7: Vocal tract profiles for nasal sounds

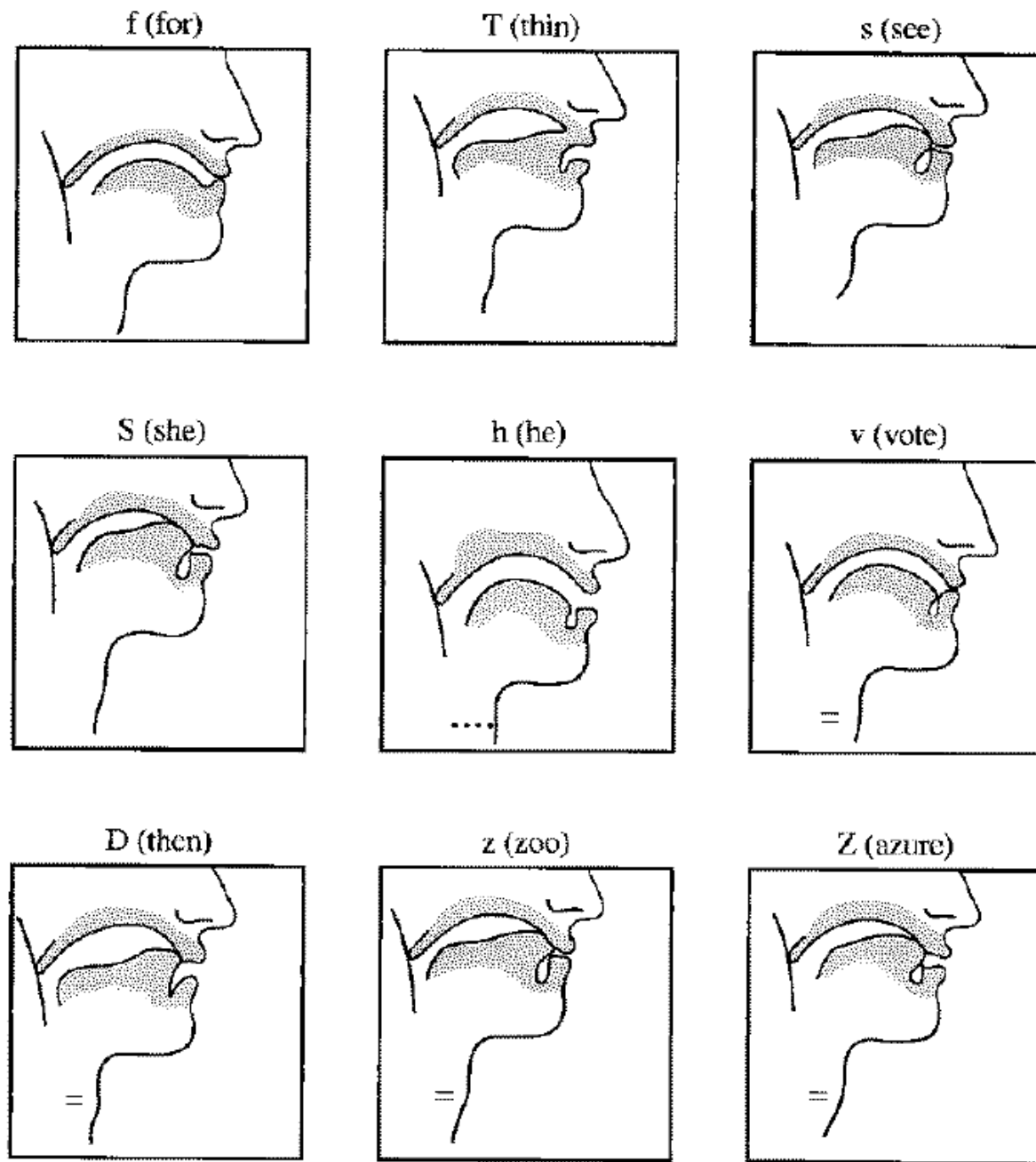


Fig 8: Vocal tract profiles for Fricative sounds

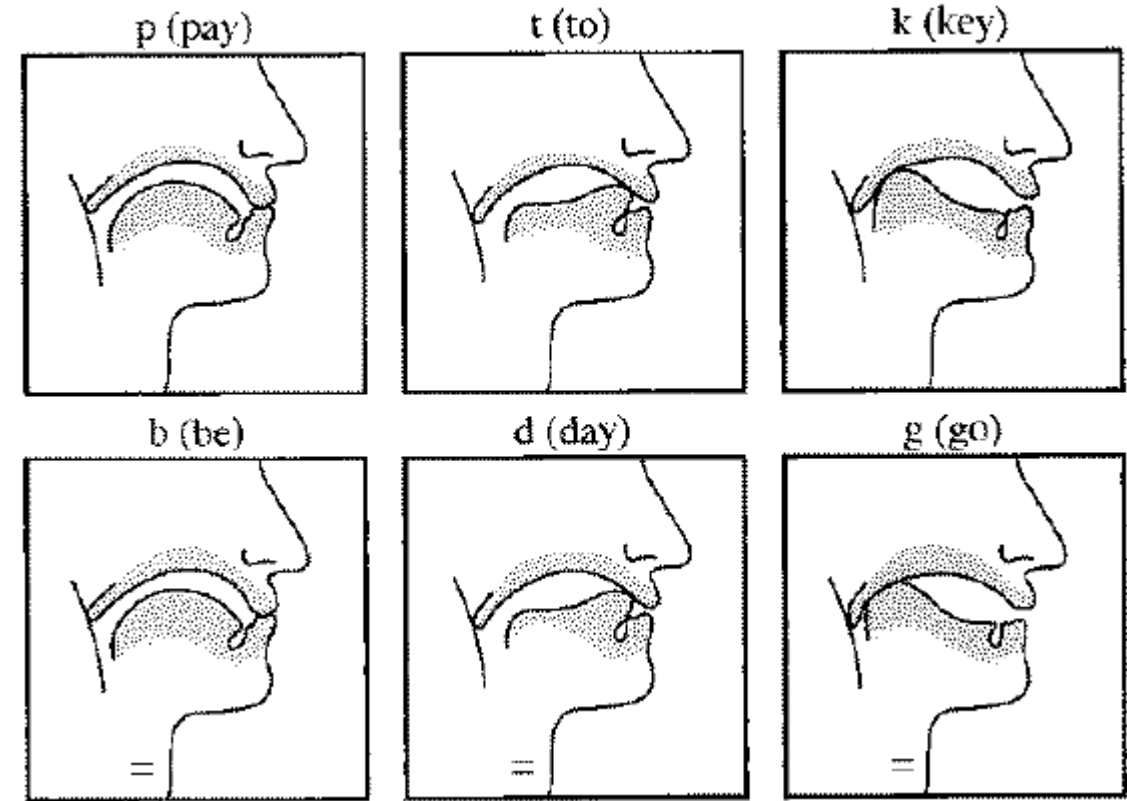
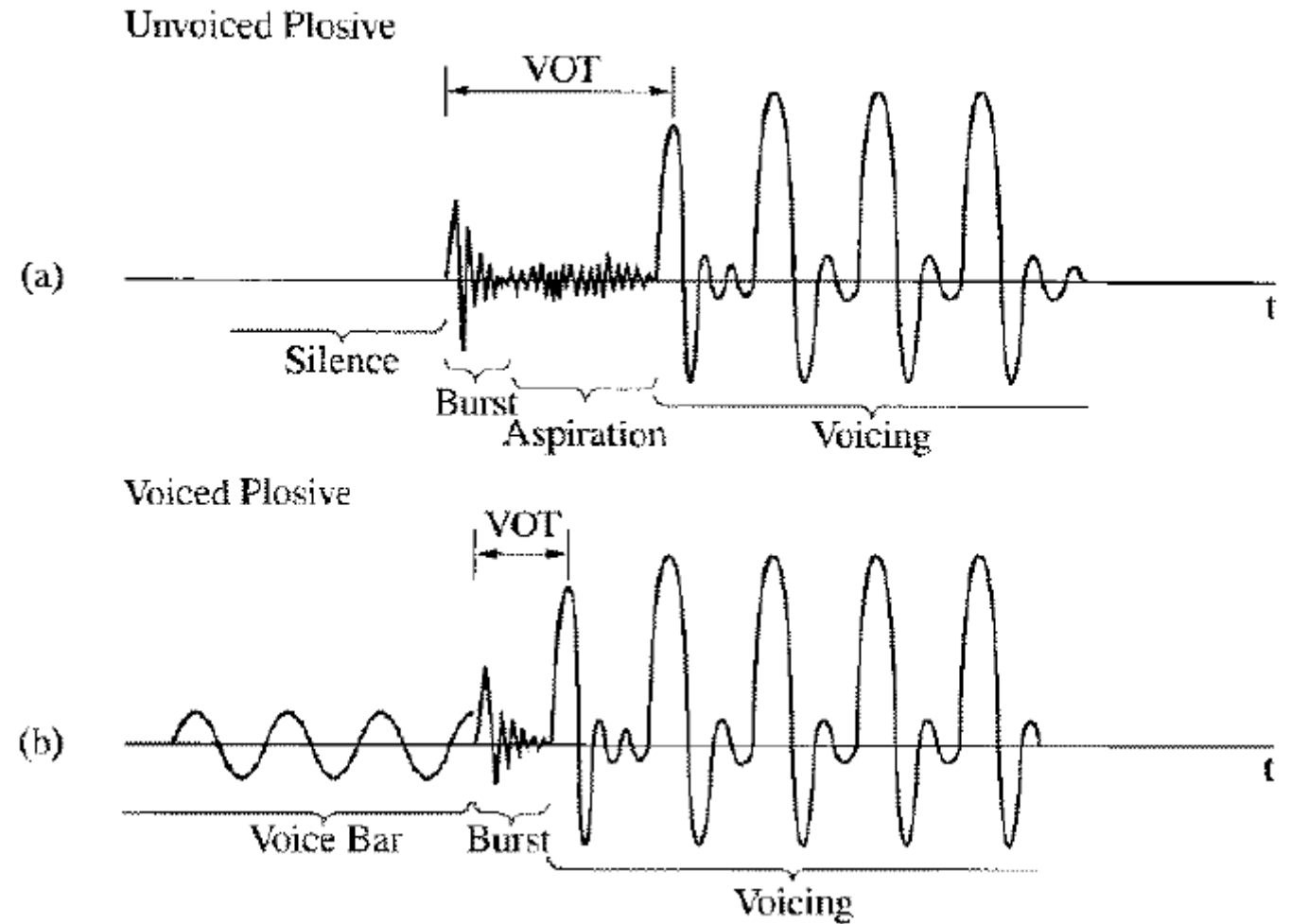


Fig 8: Vocal tract profiles for Plosive sounds

Voice Onset Time (VOT)

- Plosive is followed by a vowel
- Hence, a burst
- VOT – the time taken between the burst and the voicing onset
- Unvoiced VOT = burst + aspiration
- Voiced VOT = burst



Categories of Sound Linguistically (Contd.)

- Fundamental frequency – determine the source of sound & speaker
- Formants – determine the sound (vowel/consonant)
- **Note:** Determining vowel is easier than consonants because:
 - i. Formants are sufficient to determine a vowel
 - ii. Consonants – formants + formant transition + voicing/unvoicing + relative timing + VOT

English phonemes and corresponding features

Phoneme (IPA symbols)	Manner of articulation	Place of articulation	Voiced?	Example word
i	vowel	high front tense	yes	beat
ɪ	vowel	high front lax	yes	bit
e	vowel	mid front tense	yes	bait
ɛ	vowel	mid front lax	yes	bet
æ	vowel	low front tense	yes	bat
ɑ	vowel	low back tense	yes	cot
ɔ	vowel	mid back lax rounded	yes	caught
o	vowel	mid back tense rounded	yes	coat
ʊ	vowel	high back lax rounded	yes	book
u	vowel	high back tense rounded	yes	boot
ʌ	vowel	mid back lax	yes	but
ɜ	vowel	mid tense (retroflex)	yes	curt
ə	vowel	mid lax (schwa)	yes	about
aj (aɪ)	diphthong	low back → high front	yes	bite
ɔj (ɔɪ)	diphthong	mid back → high front	yes	boy
aw (aʊ)	diphthong	low back → high back	yes	bout
j	glide	front unrounded	yes	you
w	glide	back rounded	yes	wow
l	liquid	alveolar	yes	lull
r	liquid	retroflex	yes	roar

English phonemes and corresponding features (Contd.)

m	nasal	labial	yes	maim
n	nasal	alveolar	yes	none
ŋ	nasal	velar	yes	bang
f	fricative	labiodental	no	fluff
v	fricative	labiodental	yes	valve
θ	fricative	dental	no	thin
ð	fricative	dental	yes	then
s	fricative	alveolar strident	no	sass
z	fricative	alveolar strident	yes	zoos
ʃ	fricative	palatal strident	no	shoe
ʒ	fricative	palatal strident	yes	measure
h	fricative	glottal	no	how
p	stop	labial	no	pop
b	stop	labial	yes	bib
t	stop	alveolar	no	tot
d	stop	alveolar	yes	did
k	stop	velar	no	kick
g	stop	velar	yes	gig
tʃ	affricate	alveopalatal	no	church
dʒ	affricate	alveopalatal	yes	judge

Acoustic Characteristic of Speech

1. Phoneme Classification:

- Voiced/Unvoiced
- Vowels/Consonants

2. Time-domain Characteristics

- Amplitude
- Duration
- Energy

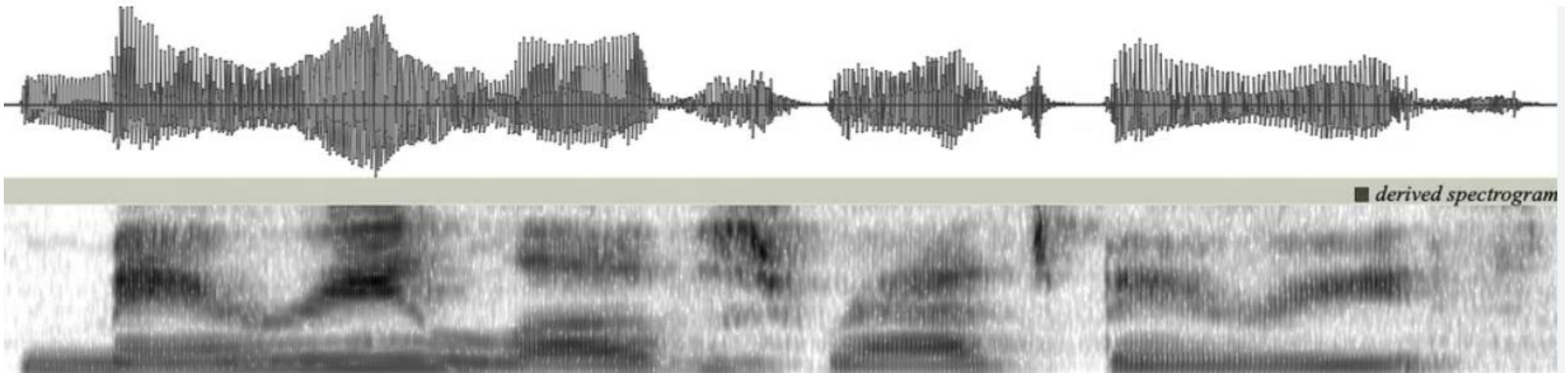
3. Frequency-domain Characteristics

- Fundamental frequency
- Formants

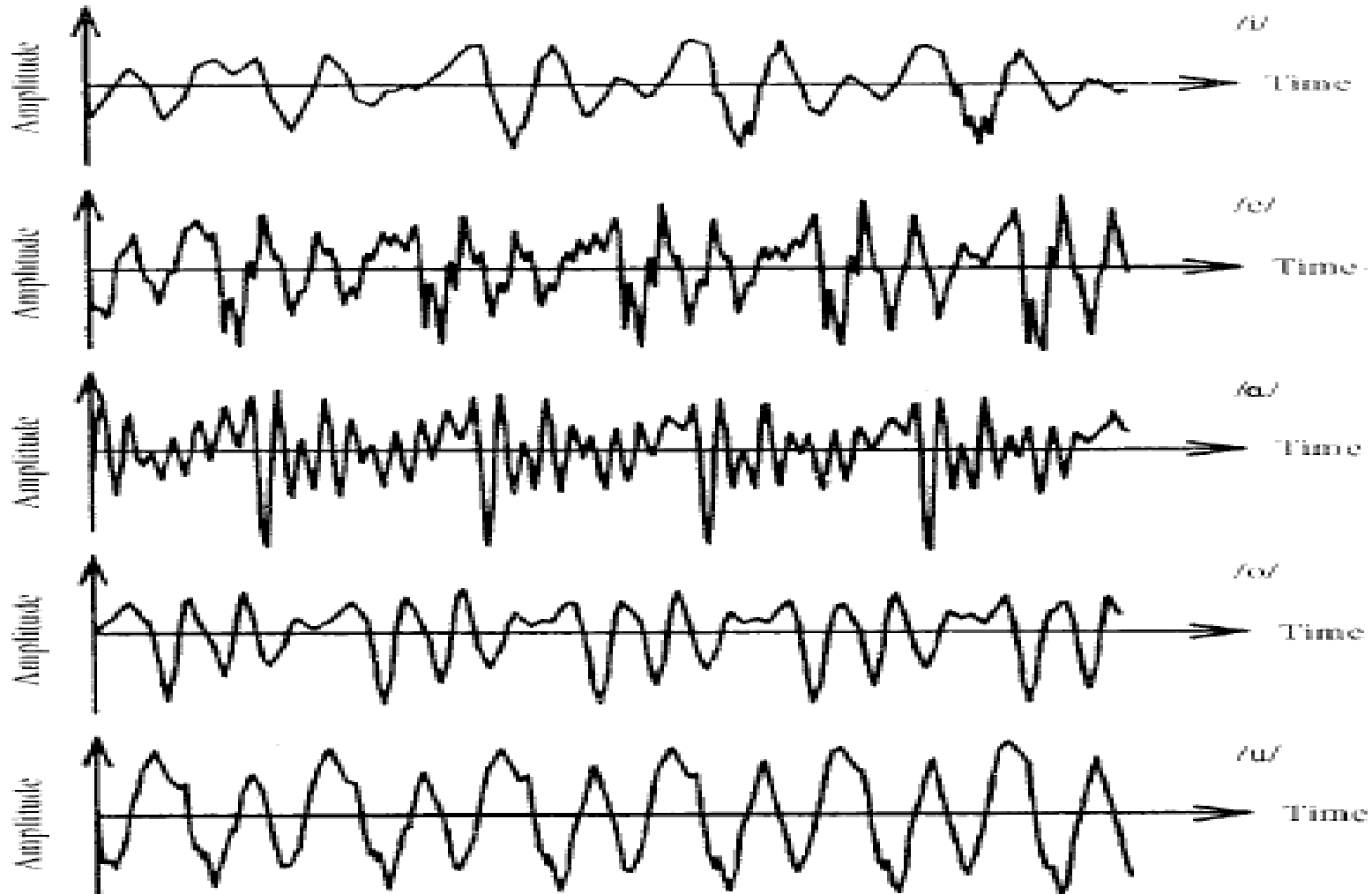
Spectrogram

Three dimensional pattern

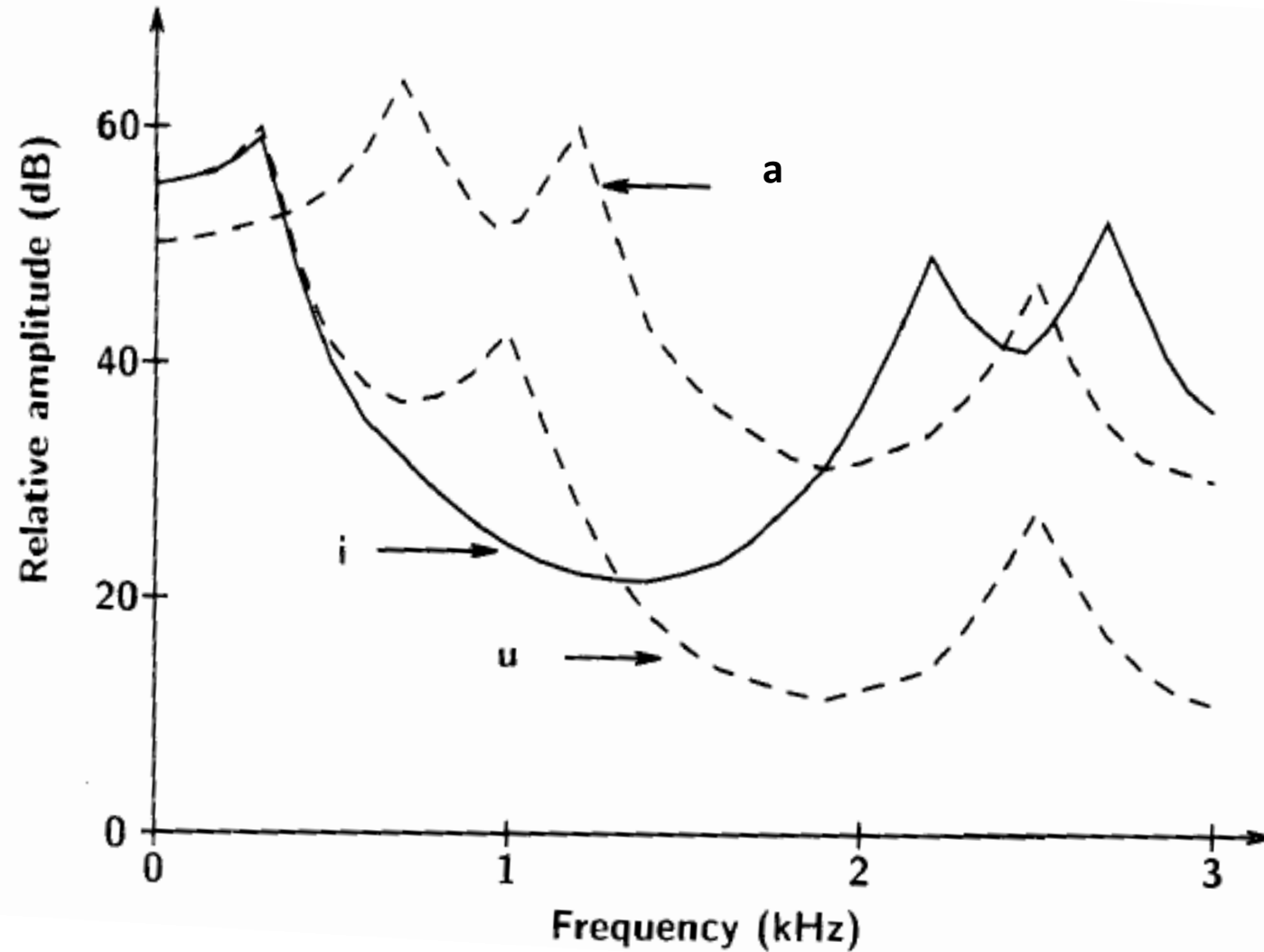
1. Horizontal axis – time
2. Vertical axis – frequency
3. Darkness of the display - amplitude



Acoustic waveforms of English vowels



Spectra of three basic vowels /a/, /i/ and /u/

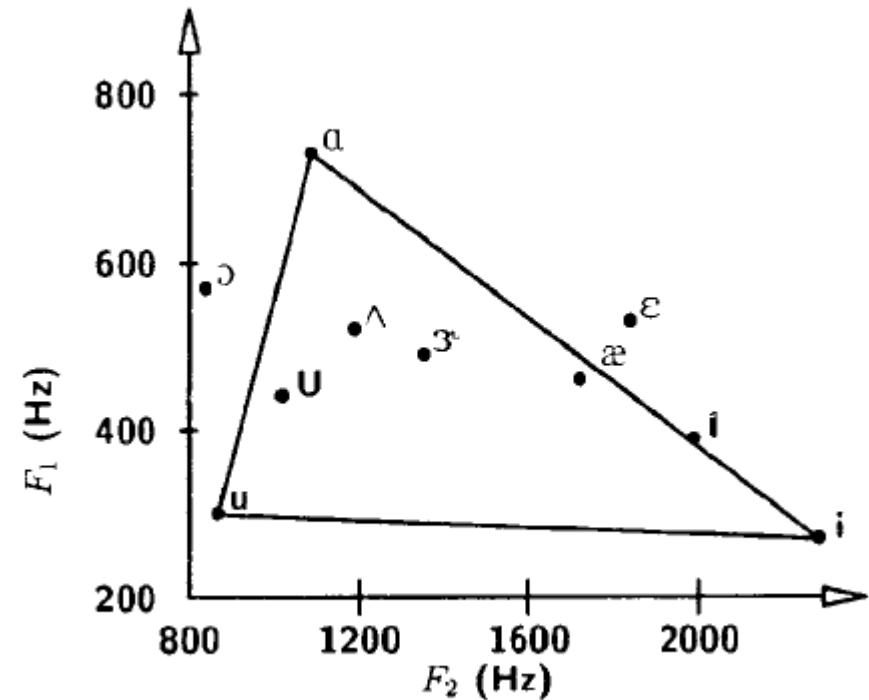
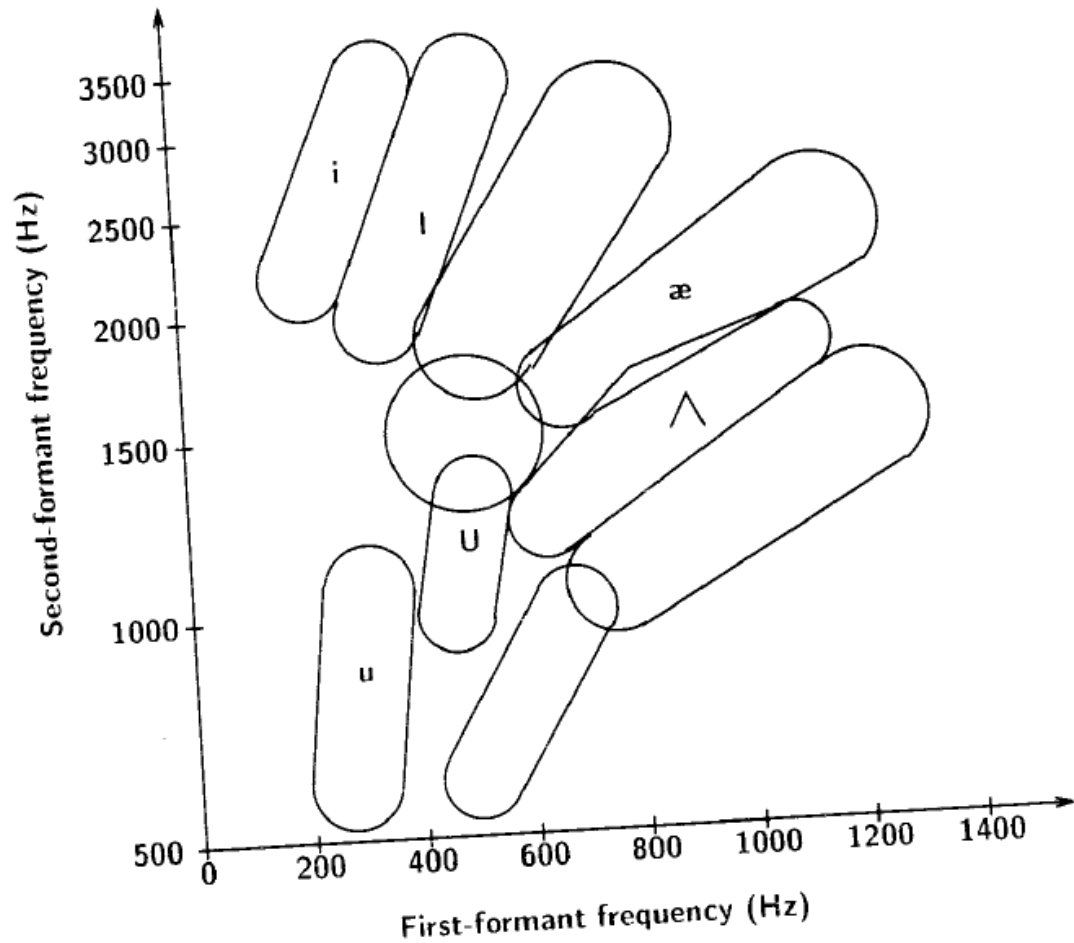


Average formant frequencies (Hz) for English vowels

- F1 relates to height of the tongue – decreases with increase in height of the tongue
- F2 relates to backness of the tongue – decreases as tongue shifts backward
- Survey: Number of male speakers: 32; Number of female speakers: 28

		i	I	e	æ	a	/ɔ/	/U/	/u/	/ʌ	ʊ
F1	male	270	390	530	660	730	570	440	300	640	490
	female	310	430	610	860	850	590	470	370	760	500
F2	male	2290	1990	1840	1720	1090	840	1020	870	1190	1350
	female	2790	2480	2330	2050	1220	920	1160	950	1400	1640
F3	male	3010	2550	2480	2410	2440	2410	2240	2240	2390	1690
	female	3310	3070	2990	2850	2810	2710	2680	2670	2780	1960

Plot of F1 vs. F2 for vowels using 60 speakers



Practical 2

1. Determining the Speaker from the Fundamental Frequency of a recorded speech:

- i. Read the recorded speech signal.
- ii. Take the vowel portion from the speech.
- iii. Find the F0 from the waveform.

Thank You...